SiGMaR

Subtype Identification for Genomics and Molecular Research

Overview, Workflow and a Guided Test Case

Gemelli Computational Biology and Bioinformatics Lab
User Documentation

1. What is SiGMaR

SiGMaR is a web application for molecular subtyping of tumor gene-expression data. A user uploads a gene expression (counts) matrix, selects one of 11 supported tumor types, and SiGMaR runs a panel of independent classification methods on the same samples, then combines their calls into a single consensus molecular subtype per sample, together with an agreement score that reflects how strongly the methods concur.

It is developed by the Gemelli Computational Biology and Bioinformatics Lab and is distributed as a self-contained desktop installer (Windows and macOS) built on Docker, so it runs locally without needing a shared server or an internet connection once installed.

2. Supported Tumor Types

SiGMaR currently supports 11 tumor types, each with its own set of reference centroids and tumor-specific classifiers:

Code	Tumor Type
BLCA	Bladder Cancer
BRCA	Breast Cancer (PAM50)
COAD	Colon Adenocarcinoma
GBM	Glioblastoma
HNSCC	Head and Neck Squamous Cell Carcinoma
LUAD	Lung Adenocarcinoma
LUSC	Lung Squamous Cell Carcinoma
MESO	Mesothelioma
OV	Ovarian Cancer
PDAC	Pancreatic Adenocarcinoma
SARC	Sarcoma

3. How It Works

3.1 Classification methods

For a given tumor type, SiGMaR runs several independent classifiers on the uploaded expression matrix and reports every method's call side by side:

- **Tumor Molecular Pathology Toolkit (TMP) models** — aklimate, cloudforest, jadbio and skgrid, four independently trained machine-learning classifiers.
- **PCE (Pearson / Cosine / Euclidean)** — nearest-centroid classification using three different distance/similarity metrics against reference subtype centroids.
- **ssGSEA** — single-sample Gene Set Enrichment Analysis against subtype-defining gene sets.
- **Nearest Template Prediction (NTP)** — template-matching classification against per-subtype marker gene templates.
- **Tumor-specific tools** — additional classifiers available only for certain tumor types (e.g. consensusOV for ovarian cancer, subscope for others).

SiGMaR then derives a **consensus subtype** for each sample — the subtype most methods agree on — plus an **agreement score** (the share of methods that voted for that call), which is used downstream to flag or filter low-confidence samples.

3.2 End-to-end workflow

1. **Start a run** — Open Subtyping and name the run.
2. **Pick the tumor type** — Determines which centroids, tumor-specific tools and TMP models are used.
3. **Upload data** — A counts file (required) and, optionally, a clinical file. Select the gene column (counts) and the sample-ID column (clinical).
4. **Normalize (optional)** — Normalize raw counts with DESeq2. Only needed if the input matrix contains raw counts; a clinical variable can be modeled to account for confounding.
5. **Run Subtyping** — SiGMaR executes all methods in sequence and computes the consensus call and agreement score.

6. **Review results** — Open Results to inspect per-method predictions, the consensus subtype, and (if clinical data was supplied) run Clinical Association to test subtype–clinical-variable relationships.

Every run is saved to disk as a study and can be reopened, re-inspected or downloaded at any time; studies can also be deleted permanently from the Results page.

3.3 Clinical Association

Once a Subtyping run has completed, its consensus calls can be tested against clinical or biological variables (e.g. stage, sex, survival status): pick the reference study, attach a clinical file, choose the variable(s) to test, optionally restrict the analysis to samples above an agreement-score threshold (default 70%), and run the association.

4. Input Files

File	Requirement	Format
Counts matrix	Required	CSV / TSV / TXT / XLSX. One column of gene identifiers, one numeric column per sample. SiGMaR auto-detects the gene column.
Clinical file	Optional (required for Clinical Association)	CSV / TSV / TXT / XLSX. One sample-ID column plus one or more clinical/biological variable columns.

Sample IDs in the counts file (column headers) and the clinical file (sample-ID column) must match exactly.

5. Results & Outputs

Every run opens as a report with three tabs:

- **Table** — per-sample predictions from every method, the consensus subtype, and the agreement score.
- **Graphics** — Subtype View (distribution of samples across predicted subtypes) and Sample View (per-method calls per sample, color-coded, so disagreement stands out).
- **Clinical Associations** — statistical tests and plots relating subtypes to clinical variables, once a clinical file is attached.

The full report can be exported as CSV, PDF, HTML or a ZIP archive; individual charts can be downloaded separately as PNG, TIFF or PDF at 150 or 300 dpi.

6. Guided Test Case

This section walks through a complete Subtyping run using a small synthetic dataset, purely to verify that a fresh SiGMaR installation starts, accepts data and produces a report end to end. The values are randomly generated (fixed seed) — they exercise the pipeline but carry no biological meaning, so the resulting subtype calls should not be interpreted.

6.1 Test data

Two files are provided alongside this guide:

File	Contents
`dummy_counts.csv`	25 genes x 12 samples (SAMPLE_01 ... SAMPLE_12), column `gene_id` + one numeric column per sample.
`dummy_clinical.csv`	12 samples with columns: `sample_id, age, sex, tumor_stage, os_time_days, os_status`.

Both column names (`gene_id` , `sample_id`) match SiGMaR's automatic-detection rules, so the app should pre-select the right column for each file without manual adjustment.

6.2 Steps

1. **Launch SiGMaR** — Start the app from the desktop shortcut created by the installer and wait for it to open in the browser.
2. **Start a run** — Go to Subtyping, give the run a name (e.g. "test_run_01").
3. **Pick a tumor type** — Select any tumor type from the list (e.g. Ovarian Cancer (OV)) — the dataset is synthetic, so the specific choice does not matter for this connectivity test.
4. **Upload the counts file** — Upload `dummy_counts.csv` . Confirm the gene column is set to `gene_id` .
5. **Upload the clinical file** — Upload `dummy_clinical.csv` . Confirm the sample-ID column is set to `sample_id` .
6. **Skip normalization** — The counts are already small random integers; normalization can be skipped or run for a full pipeline check.
7. **Run Subtyping** — Click Run and wait for all methods to complete.
8. **Check the results** — Open Results: the Table tab should list all 12 samples with a prediction from each method, a consensus subtype and an agreement score; the Graphics tab should render Subtype View and Sample View without errors.
9. **Run Clinical Association (optional)** — From Clinical Association, pick this run as the reference, choose a variable such as `tumor_stage` or `os_status` , and confirm the analysis runs and returns statistics/plots.

6.3 Expected outcome

A successful test means: the run completes without errors, all 12 samples appear in the report, every method returns a call, and the report can be exported (CSV/HTML/PDF). This confirms the installation (Docker images, app, and dependencies) is working correctly. The actual subtype assignments are not meaningful, since the input is random data.

For details on every field and page, see the in-app Documentation and About pages.